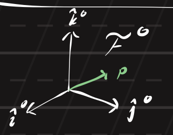
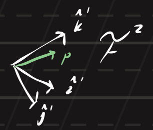
Chapter 2





$\hat{i}^i, \hat{j}^i, \hat{k}^i$ ($\cdot^i$ denotes frame)
mutually orthogonal
bases

There are different frames and we want to express a vector in one frame using the other

In $\mathcal{F}^0$ we have $\vec{p} = p_x^0 \hat{i}^0 + p_y^0 \hat{j}^0 + p_z^0 \hat{k}^0$
In $\mathcal{F}^1$ we have $\vec{p} = p_x^1 \hat{i}^1 + p_y^1 \hat{j}^1 + p_z^1 \hat{k}^1$

$$\Rightarrow \quad p_x^0 \underline{\hat{i}}^0 + p_y^0 \underline{\hat{j}}^0 + p_z^0 \underline{\hat{k}}^0 = p_x^1 \underline{\hat{i}}^1 + p_y^1 \underline{\hat{j}}^1 + p_z^1 \underline{\hat{k}}^1$$

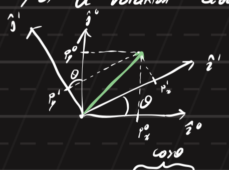
Doing a dot prod on each side with the bases of $\mathcal{F}^1$:

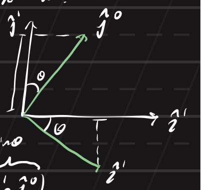
$$p_x^1 \underbrace{\hat{i}^1 \cdot \hat{i}^1}_{1} + p_y^1 \underbrace{\hat{j}^1 \cdot \hat{i}^1}_{0} + \ldots = (p_x^0 \hat{i}^0 + \ldots)\, \hat{i}^1$$

going from frame 0 to 1

$$\Rightarrow \vec{p}^{\,1} = \begin{bmatrix} p_x^1 \\ p_y^1 \\ p_z^1 \end{bmatrix} = \begin{bmatrix} \hat{i}^1 \cdot \hat{i}^0 & \hat{i}^1 \cdot \hat{j}^0 & \hat{i}^1 \cdot \hat{k}^0 \\ \hat{j}^1 \cdot \hat{i}^0 & \hat{j}^1 \cdot \hat{j}^0 & \hat{j}^1 \cdot \hat{k}^0 \\ \hat{k}^1 \cdot \hat{i}^0 & \hat{k}^1 \cdot \hat{j}^0 & \hat{k}^1 \cdot \hat{k}^0 \end{bmatrix} \begin{bmatrix} p_x^0 \\ p_y^0 \\ p_z^0 \end{bmatrix} \Rightarrow \vec{p}^{\,1} = R_0^1 \vec{p}^{\,0}$$

For a rotation about $\hat{k}^0 = \hat{k}^1$:





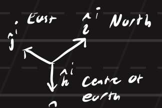
Basically projecting the original basis onto the new basis gives the factors

$$\Rightarrow p_x^1 = p_x^0 \overbrace{(\hat{i}^1 \cdot \hat{i}^0)}^{\cos\theta} + p_y^0 \overbrace{(\hat{i}^1 \cdot \hat{j}^0)}^{\sin\theta}$$

$$p_y^1 = p_x^0 \underbrace{(\hat{j}^1 \cdot \hat{i}^0)}_{-\sin\theta} + p_y^0 \underbrace{(\hat{j}^1 \cdot \hat{j}^0)}_{\cos\theta}$$

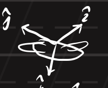
Frames:

$\mathcal{F}^i$: inertial, earth axes



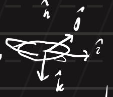
NED

$\mathcal{F}^v$: Vehicle, adjusts for pos.



aligned v
NED

$\mathcal{F}^{v1}$: vehicle frame 1, adjusts for heading



rotation about $\hat{k}^v$ by heading $\Rightarrow \vec{p}^{\,v1} = R_v^{v1}(\psi)\vec{p}^{\,v} = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \vec{p}^{\,v}$

$\mathcal{F}^{v2}$: vehicle frame 2, adjust for pitch ...

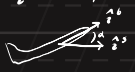
rotation about $\hat{j}^{v2} \Rightarrow p^{v2} = R_{v_1}^{v2}(\theta)\vec{p}^{\,v1} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \vec{p}^{\,v1}$

$\mathcal{F}^b$: body, adjust for roll ...

rotation about $\hat{i}^{v2} \Rightarrow p^b = R_{v2}^b(\phi)\vec{p}^{\,v2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \vec{p}^{\,v2}$

$$\Rightarrow \vec{p}^{\,b} = R_{v_2}^b(\phi) R_{v_1}^{v_2}(\theta) R_v^{v_1}(\psi)\, \vec{p}^{\,v}$$

$\mathcal{F}^s$: Stability, $\hat{i}$ is "aligned" with flow, for angle of attack



$\mathcal{F}^w$: wind, for sideslip

Velocity:

V_g – inertial velocity (ground view)
V_a – vel relative to air
V_w – wind velocity

$$\vec{V}_a = \vec{V}_g - \vec{V}_w$$

$$\Rightarrow V_g^b = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \Big\} \text{inertial velocity in body frame}$$

$$V_w^b = \begin{bmatrix} u_w \\ v_w \\ w_w \end{bmatrix} \Big\} \text{wind velocity in body frame} = R_v^b(\phi, \theta, \psi) \begin{pmatrix} w_n \\ w_e \\ w_d \end{pmatrix}$$

$$V_a^b = \begin{bmatrix} u_r \\ v_r \\ w_r \end{bmatrix} = \begin{bmatrix} u - u_w \\ v - v_w \\ w - w_w \end{bmatrix} \Big\} \text{relative airspeed in body frame}$$

From a similar analysis to above, using the definition of V_a and rotating to the body frame:

$$\begin{bmatrix} u_r \\ v_r \\ w_r \end{bmatrix} = V_a \begin{bmatrix} \cos\alpha\cos\beta \\ \sin\beta \\ \sin\alpha\cos\beta \end{bmatrix}$$

Very important for physics

$$\Rightarrow V_A = \sqrt{u_r^2 + v_r^2 + w_r^2}$$

$$\alpha = \tan^{-1}\left(\frac{w_r}{u_r}\right)$$

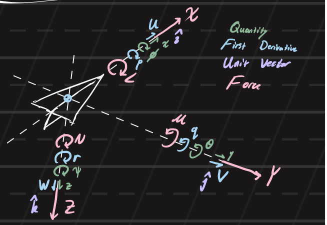
$$\beta = \sin^{-1}\left(\frac{v_r}{\sqrt{u_r^2 + v_r^2 + w_r^2}}\right)$$

Derivatives

$$\frac{d}{dt_i}\vec{p} = \frac{d}{dt_b}\vec{p} + \vec{\omega}_{b/i} \times \vec{p} \Big\} \text{angular vel of body frame relative to inertial}$$

1.) Set up simple rendering,
2.) Implement rotation functions $\Rightarrow$ one function taking angles and outputting rotation matrix
3.) Test translation and rotation

Chapter 3



Kinematic relations

each angle is in a diff FOR so we diff in that FOR and then rotate

$$\frac{d}{dt}\begin{bmatrix} p_n \\ p_e \\ p_d \end{bmatrix} = R_b^v \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \left(R_v^b\right)^T \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot\phi \\ 0 \\ 0 \end{bmatrix} + R_{v_2}^b(\phi)\begin{bmatrix} 0 \\ \dot\theta \\ 0 \end{bmatrix} + R_{v_2}^b(\phi)\, R_{v_1}^{v_2}(\theta)\begin{bmatrix} 0 \\ 0 \\ \dot\psi \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix}\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix}$$

$$\begin{bmatrix} p_n \\ p_e \\ p_d \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix}\begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Dynamics

Translation

Using $V_g^b = [u\ v\ w]^T$, the velocity with respect to ground in the body frame

$$m\frac{d\vec{V}_g}{dt_i} = \underline{f}$$

we want in body frame

$$\Rightarrow = \frac{d\vec{V}_g}{dt_b} + \vec{\omega}_{b/i} \times \vec{V}_g$$

from earlier $\vec{\omega}_{b/i}$ is $[\dot\phi, \dot\theta, \dot\psi]^T$

Alternatively:

$$m\left(\frac{d\vec{V}_g^b}{dt_b} + \vec{\omega}_{b/i}^b \times \vec{V}_g^b\right) = \vec{f}^b$$

which is in the body frame, $V_g^b = [u, v, w]^T$, $W_{b/i}^b = [p, q, r]$

$$\frac{d\vec{V}_g}{dt_b} = \begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} \Rightarrow \begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} = \begin{bmatrix} rv - qw \\ pw - ru \\ qu - pv \end{bmatrix} + \frac{1}{m}\begin{bmatrix} f_x^b \\ f_y^b \\ f_z^b \end{bmatrix}$$

Rotational Motion

$$\frac{d\vec{L}^i}{dt_i} = \vec{\tau} = \frac{d\vec{L}}{dt_b} + \vec{\omega}_{b/i} \times \vec{L}$$

$\vec{L}^i$ ← angular momentum

Assuming moments taken about COM

Expressing in body frame:

$$\frac{d\vec{L}^b}{dt_b} + \vec{\omega}_{b/i}^b \times \vec{L}^b = \vec{\tau}$$

$$\underline{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ \cdot & I_{yy} & I_{yz} \\ \cdot & \cdot & I_{zz} \end{bmatrix}, \quad \frac{d\underline{I}}{dt} = 0$$

$$\vec{L}^b = \underline{I}^b \vec{\omega}_{b/i} \Rightarrow \frac{d\vec{L}}{dt_b} = \underline{I}\frac{d\omega_{b/i}^b}{dt_b}$$

$$\Rightarrow \underline{I}\frac{d\vec{\omega}_{b/i}^b}{dt_b} + \vec{\omega}_{b/i}^b \times \left(\underline{I}\,\vec{\omega}_{b/i}^b\right) = \vec{\tau}^b$$

$$\dot{\vec{\omega}}_{b/i}^b = \begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} \text{ and } \dot{\omega}_{b/i}^b = \underline{I}^{-1}\left[-\omega_{b/i}^b \times \left(\underline{I}\,\vec{\omega}_{b/i}^b\right) + \vec{\tau}^b\right]$$

Using Symmetry:

$$\underline{I} = \begin{bmatrix} I_x & 0 & -I_{xz} \\ 0 & I_y & 0 \\ -I_{xz} & 0 & I_z \end{bmatrix} \Rightarrow \underline{I}^{-1} = \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix} \quad \text{with } \Gamma = I_xI_z - I_{xz}^2$$

$$\Rightarrow \begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} = \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix}\left(\begin{bmatrix} 0 & r & -q \\ -r & 0 & p \\ q & -p & 0 \end{bmatrix}\begin{bmatrix} I_x & 0 & -I_{xz} \\ 0 & I_y & 0 \\ -I_{xz} & 0 & I_z \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix} + \begin{bmatrix} L \\ \mathcal{M} \\ N \end{bmatrix}\right)$$

$$= \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix}\left(\begin{bmatrix} I_{xz}pq + (I_y - I_z)qr \\ I_{xz}(r^2 - p^2) + (I_z - I_x)pr \\ (I_x - I_y)pq - I_{xz}qr \end{bmatrix} + \begin{bmatrix} L \\ \mathcal{M} \\ N \end{bmatrix}\right)$$

$$= \begin{bmatrix} \frac{I_{xz}(I_x - I_y + I_z)}{I_xI_z - I_{xz}^2}pq - \frac{I_z(I_z - I_y) + I_{xz}^2}{I_xI_y - I_{xz}^2}qr + \frac{I_z}{I_xI_z - I_{xz}^2}L + \frac{I_{xz}}{I_xI_z - I_{xz}^2}N \\ \frac{I_z - I_x}{I_y}pr - \frac{I_{xz}}{I_y}(p^2 - r^2) + \frac{\mathcal{M}}{I_y} \\ \frac{(I_x - I_y)I_x + I_{xz}^2}{I_xI_y - I_{xz}^2}pq - \frac{I_{xz}(I_x - I_y + I_z)}{I_xI_z - I_{xz}^2}qr + \frac{I_{xz}}{I_xI_z - I_{xz}^2}L + \frac{I_x}{I_xI_z - I_{xz}^2}N \end{bmatrix}$$

$$= \begin{bmatrix} \frac{I_{xz}(I_x - I_y + I_z)}{I_xI_z - I_{xz}^2}pq - \frac{I_z(I_z - I_y) + I_{xz}^2}{I_xI_y - I_{xz}^2}qr \\ \frac{I_z - I_x}{I_y}pr - \frac{I_{xz}}{I_y}(p^2 - r^2) \\ \frac{(I_x - I_y)I_x + I_{xz}^2}{I_xI_y - I_{xz}^2}pq - \frac{I_{xz}(I_x - I_y + I_z)}{I_xI_z - I_{xz}^2}qr \end{bmatrix} + \begin{bmatrix} \frac{I_z}{I_xI_z - I_{xz}^2}L + \frac{I_{xz}}{I_xI_z - I_{xz}^2}N \\ \frac{\mathcal{M}}{I_y} \\ \frac{I_{xz}}{I_xI_z - I_{xz}^2}L + \frac{I_x}{I_xI_z - I_{xz}^2}N \end{bmatrix}$$

1.) we can use $\begin{bmatrix} p_n \\ p_e \\ p_d \end{bmatrix} = \ldots$, $\begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} = \ldots$, $\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \ldots$ and $\begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} = \ldots$ to represent our aircraft

2.) Implement the different formulations using both hand calcs and the full vector equations

Re-expressing with Quaternions

Quaternion $\underline{e} \in \mathbb{R}^4$, $\underline{e} = e_0 + e_1\hat{i} + e_2\hat{j} + e_3\hat{k}$, $\left.\hat{i}^2 = \hat{j}^2 = \hat{k}^2 = \hat{i}\hat{j}\hat{k} = -1 \right\}$ from this $\hat{i}\hat{j} = \hat{k} = -\hat{j}\hat{k}$

$\Rightarrow$ equivalently $\underline{e} = [e_0, \vec{\varepsilon}]^T$, $\underline{\varepsilon} = [e_1, e_2, e_3]^T$

If you rotate by θ about an axis $\vec{v}$, the quaternion is related via:

$$e_0 = \cos\left(\frac{\theta}{2}\right), \quad \underline{\varepsilon} = \vec{v}\sin\left(\frac{\theta}{2}\right) = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

A quaternion product is defined as:

$$\underline{a} \otimes \underline{b} = \begin{bmatrix} a_0 b_0 - \vec{\alpha}\cdot\vec{\beta} \\ a_0\vec{\beta} + b_0\vec{\alpha} + \underbrace{\vec{\alpha}\times\vec{\beta}}_{\text{makes it non commutative}} \end{bmatrix}, \quad \text{with } \underline{a} = [a_0, \vec{\alpha}]^T, \ \underline{b} = [b_0, \vec{\beta}]^T$$

To actually rotate a vector $\vec{r}$, it's expressed as $(0, \vec{r})$, and we sandwich between $\overbrace{e \text{ and its conjugate } e^*}^{\left(\text{hence } \frac{\theta}{2} \text{ as its done twice}\right)}$ $(= [e_0, -\vec{\varepsilon}]^T)$

$$(0, \vec{r}^{\,v}) = \underline{e} \otimes (0, \vec{r}^{\,b}) \otimes \underline{e}^*, \quad (0, \vec{r}^{\,b}) = \underline{e}^* \otimes (0, \vec{r}^{\,v}) \otimes \underline{e}$$

$\Rightarrow$ $\underline{e} \otimes (\cdot) \otimes \underline{e}^*$ takes body components to vehicle components

$\underline{e}$ is the attitude of the body frame relative to vehicle (ie inertial without translation)

The rotation Matrix

expanding:

$$\underline{e} \otimes (0, \vec{r}) \otimes \underline{e}^* = \left[-\vec{\varepsilon}\cdot\vec{r}^{\,b}, \ e_0\vec{r}^{\,b} + \vec{\varepsilon}\times\vec{r}^{\,b}\right]^T \otimes [e_0, -\vec{\varepsilon}]^T = \vec{r}^{\,v}$$

$$= \left[-e_0(\vec{\varepsilon}\cdot\vec{r}^{\,b}) + (e_0\vec{r} + \vec{\varepsilon}\times\vec{r}^{\,b})\cdot\vec{\varepsilon}, \ (\vec{\varepsilon}\cdot\vec{r}^{\,b})\vec{\varepsilon} + e_0(e_0\vec{r} + \vec{\varepsilon}\times\vec{r}^{\,b}) + (e_0\vec{r} + \vec{\varepsilon}\times\vec{r}^{\,b})\times(-\vec{\varepsilon})\right]^T$$

and with $\vec{\varepsilon}\times(\vec{\varepsilon}\times\vec{r}) = \vec{\varepsilon}(\vec{\varepsilon}\cdot\vec{r}) - \vec{r}(\vec{\varepsilon}\cdot\vec{\varepsilon})$ (vector identity, very common when manipulating Navier-Stokes)

$$\Rightarrow \left[\underbrace{-e_0(\vec{\varepsilon}\cdot\vec{r}) + e_0(\vec{r}\cdot\vec{\varepsilon})}_{\text{equal but opposite sign}} + \underbrace{\overset{0}{\cancel{(\vec{\varepsilon}\times\vec{r})\cdot\vec{\varepsilon}}}}_{\substack{\varepsilon\times\vec{r} \text{ is } \perp \\ \text{to both } \vec{\varepsilon} \text{ and } \vec{r}, \\ \therefore \text{ dot} = 0}}, \ (\vec{\varepsilon}\cdot\vec{r})\vec{\varepsilon} + e_0^2\vec{r} + \underbrace{e_0(\vec{\varepsilon}\times\vec{r}) - e_0(\vec{r}\times\vec{\varepsilon})}_{\substack{\vec{r}\times\vec{\varepsilon} = -\vec{\varepsilon}\times\vec{r} \\ = 2e_0(\vec{\varepsilon}\times\vec{r})}} \underbrace{-(\vec{\varepsilon}\times\vec{r})\times\vec{\varepsilon}}_{\substack{= +\vec{\varepsilon}\times(\vec{\varepsilon}\times\vec{r}) \\ = (\vec{\varepsilon}\cdot\vec{r})\vec{\varepsilon} - \underbrace{(\vec{\varepsilon}\cdot\vec{\varepsilon})}_{\varepsilon^T\varepsilon}\vec{r}}}\right]^T$$

$$\Rightarrow \vec{r}^{\,v} = \underbrace{(e_0^2 - \vec{\varepsilon}^T\vec{\varepsilon})\vec{r}^{\,b}}_{(e_0^2 - \vec{\varepsilon}^T\vec{\varepsilon})\underline{I}} + \underbrace{2(\vec{\varepsilon}\cdot\vec{r}^{\,b})\vec{\varepsilon}}_{\substack{\text{rearranging + associativity} \\ \to 2(\varepsilon\varepsilon^T)}} + 2e_0(\vec{\varepsilon}\times\vec{r}^{\,b})$$

$$\vec{\varepsilon}\times\vec{r} = \begin{bmatrix} e_2 r_3 - e_3 r_2 \\ e_3 r_1 - e_1 r_3 \\ e_1 r_2 - e_2 r_1 \end{bmatrix} = \begin{bmatrix} 0 & -e_3 & e_2 \\ e_3 & 0 & -e_1 \\ -e_2 & e_1 & 0 \end{bmatrix}\vec{r}$$

$$\Rightarrow R_b^v(\underline{e}) = \begin{bmatrix} e_0^2 + e_1^2 - e_2^2 - e_3^2 & 2(e_1e_2 - e_0e_3) & 2(e_1e_3 + e_0e_2) \\ 2(e_1e_2 + e_0e_3) & e_0^2 - e_1^2 + e_2^2 - e_3^2 & 2(e_2e_3 - e_0e_1) \\ 2(e_1e_3 - e_0e_2) & 2(e_2e_3 + e_0e_1) & e_0^2 - e_1^2 - e_2^2 + e_3^2 \end{bmatrix} = R_v^b(e)^T$$

Quaternion Kinematics

Over interval dt, body will turn by angle $\|\vec{\omega}\|dt$ about axis $\frac{\vec{\omega}}{\|\vec{\omega}\|}$ (in body axes as $\vec{\omega} = [p, q, r]^T$)

With $\underline{e} = [\cos\frac{\theta}{2}, \vec{v}\sin(\frac{\theta}{2})]$

$\Rightarrow \delta\underline{e} \approx [1, \frac{1}{2}\vec{\omega}\,dt]$

rotating about current body axes will compose on the right

$\Rightarrow \underline{e}(t + dt) = \underline{e}(t) \otimes \delta\underline{e} = \underline{e}(t) + \underline{e}(t) \otimes (0, \frac{1}{2}\vec{\omega})dt$

$\Rightarrow \dot{\underline{e}} = \frac{1}{2}\underline{e} \otimes (0, \vec{\omega}) = \frac{1}{2}[0 - \vec{\varepsilon}\cdot\vec{\omega}, \ e_0\vec{\omega} + \vec{\varepsilon}\times\vec{\omega}]$

$$\Rightarrow \begin{bmatrix} \dot{e}_0 \\ \dot{e}_1 \\ \dot{e}_2 \\ \dot{e}_3 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix}\begin{bmatrix} e_0 \\ e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

↑

the magnitude will change, so we need to renormalise (or do some gradient descent lets see)

To sum up

The state is composed of:

- position (inertial frame) $= [p_n, p_e, p_d]^T = \vec{p}$
- ground velocity (body frame) $= [u, v, w]^T = \vec{v}_b$
- attitude (from body to vehicle) $= [e_0, e_1, e_2, e_3]^T = \underline{e} = [e_0, \vec{\varepsilon}]$, $\vec{\varepsilon} = [e_1 \ e_2 \ e_3]^T$
- rates (body frame) $= [p, q, r]^T = \vec{\omega}_b$

For forces and moments on the craft, we use body frame (usually)

1.) Calculate the rate of change of velocity and rates in the body frame
2.) Use the kinematic relations to express the position in the vehicle frame using integrated body-frame velocity, and to update the attitude based on the body-frame rates.

Dynamics (body-frame)

$$\dot{\vec{v}}_b = -\vec{\omega}_b \times \vec{v}_b + \frac{1}{m}\vec{f}_b$$

$$\dot{\vec{\omega}}_b = \underline{I}^{-1}(\vec{\tau}_b - \vec{\omega}_b \times \underline{I}\vec{\omega}_b)$$

Kinematic (body to vehicle frame)

$$\dot{\vec{p}} = \underline{R}_b^v(e)\,\vec{v}_b$$

$$\dot{e} = \frac{1}{2}(\underline{e} \otimes [0, \vec{\omega}_b])$$

Forces and Moments

Forces and moments can be split (aircraft also sometimes include an augmented mass matrix)

$$\vec{f}_b = \overbrace{f_g}^{\text{gravity}} + \overbrace{f_a}^{\text{aero}} + \overbrace{f_p}^{\text{propulsive}}$$

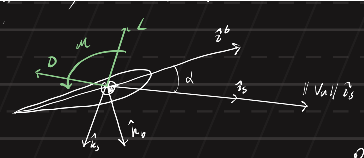
$$\vec{\tau}_b = \underbrace{\tau_a}_{\text{aero}} + \underbrace{\tau_p}_{\text{propulsive}}$$

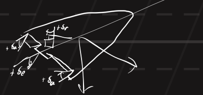
Gravity

$$\vec{f}_g^{\,v} = [0 \;\; 0 \;\; mg]$$

$$\Rightarrow \vec{f}_g^{\,b} = R_v^b(e)\,\vec{f}_g^{\,b}$$

Aerodynamic Forces and Moments





Pos deflection based on pos force dir using RHR

Diff aircraft have diff control surfaces but they can always be expressed as something like the standard aileron-elevator setup

$$F_{lift} = \tfrac{1}{2}\rho V_a^2 S_{ref}\, C_L(\alpha, q, \delta_e)$$
$$F_{drag} = \tfrac{1}{2}\rho V_a^2 S_{ref}\, C_D(\alpha, q, \delta_e)$$
$$m = \tfrac{1}{2}\rho V_a^2 S_{ref}\, c\, C_m(\alpha, q, \delta_e)$$

Longitudinal

$$F_{lift} = \tfrac{1}{2}\rho V_a^2 S_{ref}\left[C_{L_0} + \frac{\partial C_L}{\partial \alpha}\alpha + \overbrace{\frac{\partial C_L}{\partial q}}^{\text{not non dimensional}} q + \frac{\partial C_L}{\partial \delta_e}\delta_e\right] \Rightarrow \left(\times \frac{c}{2V_a}\right) = \tfrac{1}{2}\rho V_a^2 S_{ref}\left[C_{L_0} + \frac{\partial C_L}{\partial \alpha}\alpha + \frac{\partial C_L}{\partial\left(\frac{qc}{2V_a}\right)}\underbrace{\frac{c}{2V_a}}_{\text{nondimensionalize}} q + \frac{\partial C_L}{\partial \delta_e}\delta_e\right]$$

$$F_{drag} = \tfrac{1}{2}\rho V_a^2 S_{ref}\left[C_{D_0} + \frac{\partial C_D}{\partial \alpha}\alpha + \frac{\partial C_D}{\partial\left(\frac{qc}{2V_a}\right)}\frac{c}{2V_a} q + \frac{\partial C_D}{\partial \delta_e}\delta_e\right]$$

$$m = \tfrac{1}{2}\rho V_a^2 S_{ref}\, c\left[C_{m_0} + \frac{\partial C_m}{\partial \alpha}\alpha + \frac{\partial C_m}{\partial\left(\frac{qc}{2V_a}\right)}\frac{c}{2V_a} q + \frac{\partial C_m}{\partial \delta_e}\delta_e\right]$$

Nothing accounts for stall, but a stall model (flat plate) can be spliced in ($C_{L_0} + C_{L_\alpha}\alpha = C_L(\alpha)$)

$$C_{L,\,\text{flat plate}} = 2\,\text{sign}(\alpha)\sin^2\alpha\cos\alpha \;\Big|\; \text{Blending: } \sigma(\alpha) = \frac{1 + e^{-M(\alpha-\alpha_0)} + e^{M(\alpha+\alpha_0)}}{(1+e^{-M(\alpha-\alpha_0)})(1+e^{M(\alpha+\alpha_0)})}$$

α_0 = stall transition angle

M = transition rate

$$\Rightarrow C_L(\alpha) = (1-\sigma(\alpha))\left[C_{L_0} + \frac{\partial C_L}{\partial \alpha}\alpha\right] + \sigma(\alpha)\left[2\,\text{sign}(\alpha)\sin^2(\alpha)\cos(\alpha)\right]$$

$$\Rightarrow C_D(\alpha) = (1-\sigma(\alpha))\left[C_{D_0} + \frac{\left(C_{L_0} + \frac{\partial C_L}{\partial \alpha}\alpha\right)^2}{\pi AR\, e}\right] + \sigma(\alpha)\cdot 2\sin^2(\alpha)\,|\sin(\alpha)|$$

Filling in the other half of the flat plate

$(\cdot)\cos\alpha = L \;\therefore\; (\cdot)|\sin(\alpha)| = D$ (always opposite)

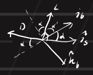
For small aircraft

$$\frac{\partial C_L}{\partial \alpha} = \frac{\pi AR}{1 + \sqrt{1 + \left(\frac{AR}{2}\right)^2}}$$

$$\Rightarrow C_m(\alpha) = C_{m_0} + \frac{\partial C_m}{\partial \alpha}\alpha$$

simpler linear model because its hard to model

Conversion to body axes is a rotation by α

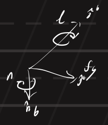


$$\Rightarrow f_x = L\sin\alpha - D\cos\alpha$$
$$f_z = -L\cos\alpha - D\sin\alpha$$
$$\Rightarrow \begin{bmatrix} f_x \\ f_z \end{bmatrix} = \begin{bmatrix} -\cos\alpha & \sin\alpha \\ -\sin\alpha & -\cos\alpha \end{bmatrix}\begin{bmatrix} D \\ L \end{bmatrix}$$

Lateral

(switching to lowercase l, m, n for moments cuz subscripts clash)

$$f_y = \tfrac{1}{2}\rho V_a^2 S_{ref}\, C_Y(\beta, p, r, \delta_a, \delta_r)$$
$$l = \tfrac{1}{2}\rho V_a^2 S_{ref}\, b\, C_l(\beta, p, r, \delta_a, \delta_r)$$
$$n = \tfrac{1}{2}\rho V_a^2 S_{ref}\, b\, C_n(\beta, p, r, \delta_a, \delta_r)$$



Same approach as before

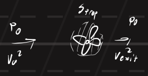
$$f_y = \tfrac{1}{2}\rho V_a^2 S\left[C_{Y_0} + \frac{\partial C_Y}{\partial \beta}\beta + \frac{\partial C_Y}{\partial\left(\frac{pb}{2V_a}\right)}\frac{b}{2V_a}p + \frac{\partial C_Y}{\partial\left(\frac{rb}{2V_a}\right)}\frac{b}{2V_a}r + \frac{\partial C_Y}{\partial \delta_a}\delta_a + \frac{\partial C_Y}{\partial \delta_r}\delta_r\right]$$

$$l = \tfrac{1}{2}\rho V_a^2 S b\left[C_{l_0} + \frac{\partial C_l}{\partial \beta}\beta + \frac{\partial C_l}{\partial\left(\frac{pb}{2V_a}\right)}\frac{b}{2V_a}p + \frac{\partial C_l}{\partial\left(\frac{rb}{2V_a}\right)}\frac{b}{2V_a}r + \frac{\partial C_l}{\partial \delta_a}\delta_a + \frac{\partial C_l}{\partial \delta_r}\delta_r\right]$$

$$n = \tfrac{1}{2}\rho V_a^2 S b\left[C_{n_0} + \frac{\partial C_n}{\partial \beta}\beta + \frac{\partial C_n}{\partial\left(\frac{pb}{2V_a}\right)}\frac{b}{2V_a}p + \frac{\partial C_n}{\partial\left(\frac{rb}{2V_a}\right)}\frac{b}{2V_a}r + \frac{\partial C_n}{\partial \delta_a}\delta_a + \frac{\partial C_n}{\partial \delta_r}\delta_r\right]$$

already in body frame

Propulsion



linear relationship between PWM command and V_{exit}

$\Rightarrow V_{exit} = k_{motor}\,\delta_t$] thrust input

$$P_{upstream} \rightarrow \;\; \Rightarrow P_{downstream}$$

$$\Rightarrow F_{xp} = S_{prop} C_{prop}(P_{downstream} - P_{upstream})$$
$$= \tfrac{1}{2}\rho S_{prop} C_{prop}\left[(k_{motor}\delta_t)^2 - V_a^2\right] \checkmark \text{ bernoulli}$$

$$\Rightarrow \vec{f}_p = \tfrac{1}{2}\rho S_{prop} C_{prop}\begin{pmatrix} (k_{motor}\delta_t)^2 - V_a^2 \\ 0 \\ 0 \end{pmatrix}$$

We assume ρ acts in $\hat{i}_b$, passing through centroid, so no moments produced

F_y produces a torque, but how much depends on Ω, and blade geometry.

$$F \propto P\cdot A \propto V^2 \propto (k_\Omega \delta_t)^2$$

$$\Rightarrow T_p = -\underset{\text{prop speed}}{k_{T_p}}(k_\Omega \delta_t)^2$$

constant for how much angular velocity changes with δ_t

$$\vec{\tau}_p = \begin{bmatrix} -k_{T_p}(k_\Omega \delta_t)^2 \\ 0 \\ 0 \end{bmatrix}$$

Atmospheric Disturbances

$$\underbrace{\vec{V}_g}_{\text{ground}} = \underbrace{\vec{V}_a}_{\text{air-relative}} + \underbrace{\vec{V}_w}_{\text{wind}}, \qquad \vec{V}_w = \underbrace{\vec{V}_{w_s}}_{\text{steady}} + \underbrace{\vec{V}_{w_g}}_{\text{stochastic wind/gust}}$$

$\vec{V}_{ws}$ is usually in inertial frame, $\vec{V}_{ws}^{\,i} = \begin{bmatrix} w_{ns} \\ w_{es} \\ w_{ds} \end{bmatrix}$

$\vec{V}_{wg}$ is usually in body frame, $\vec{V}_{wg}^{\,b} = \begin{bmatrix} u_{wg} \\ v_{wg} \\ w_{wg} \end{bmatrix}$

Stochastic Model (Dryden):

$$H_u(s) = \sigma_u\sqrt{\frac{2V_a}{L_u}}\,\frac{1}{s + \frac{V_a}{L_u}}$$
$$H_v(s) = \sigma_v\sqrt{\frac{3V_a}{L_v}}\,\frac{\left(s + \frac{V_a}{\sqrt{3}L_v}\right)}{\left(s + \frac{V_a}{L_v}\right)^2}$$
$$H_w(s) = \sigma_w\sqrt{\frac{3V_a}{L_w}}\,\frac{\left(s + \frac{V_a}{\sqrt{3}L_w}\right)}{\left(s + \frac{V_a}{L_w}\right)^2}$$

Transfer Functions

σ_i – intensity along vehicle frame axes

L_i – spatial wavelengths

Two forms: (η = unit-intensity white noise, $n_k \sim N(0,1)$)

$$H(s) = \frac{K}{s+a} \Rightarrow \dot{x} = -ax + \eta$$
$$u = Kx$$

$$H(s) = \frac{K(s+b)}{(s+a)^2}, \quad (s+a)^2 = s^2 + 2as + a^2 \;\Bigg\}\; \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -a^2 & -2a \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\eta$$
$$v_g / w_g = K\cdot(b x_1 + x_2)$$

random noise is scaled by $\sqrt{\Delta t}$, not Δt

$$\Rightarrow \dot{\vec{x}} = \underline{A}\vec{x} + \underline{B}\eta \Rightarrow x_{n+1} = \vec{x}_n + \underline{A}\vec{x}_n\Delta t + \sqrt{\Delta t}\, n_k$$

Basically get n_k, advance x, get u_g, v_g, w_g, then in the RK4 steps use $\vec{w}_b = \underline{R}^T(e)\cdot\vec{w}^{\,i}_{steady} + [u_g, v_g, w_g]^T$, and $\vec{V}_a = \vec{V}_b - \vec{w}_b$

1.) Prob subclass the Body struct to implement different aircraft (Config struct set by factory functions, and a wrench method to be overriden for each type)

2.) Have a whole file of force models which accept the params, and we override the wrench with those

3.) Maybe reorganize into a "World", "Air System", "Body System"